SIMATS ENGINEERING

Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Max Marks:100

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

Q1. Thresholding and Intensity-Level Slicing – PCB Inspection

A PCB inspection camera captures a row of pixels with the following grey levels:

r = [25, 60, 95, 130, 165, 200, 235]

The solder joints of interest lie in the intensity band 90–170. The display range is 0–255.

Tasks:

a) Apply the binary threshold transformation s = 255 if r ≥ 128, otherwise s = 0, and list the output values.

b) Apply intensity-level slicing that sets every pixel in the band 90–170 to 255 and leaves all other pixels unchanged. List the output values.

c) Calculate the number and percentage of pixels highlighted by the slicing operation.

d) Calculate the mean intensity of the original row and of the sliced output.

e) Based on the numerical results, determine which of the two transformations is more appropriate for isolating the solder joints and justify your answer.

Q2. Piecewise-Linear Contrast Stretching – X-Ray Image Enhancement

A medical X-ray image has most of its diagnostic detail between grey levels 70 and 140. A piecewise-linear transformation is defined by the control points $(r_1, s_1) = (70, 20)$ and $(r_2, s_2) = (140, 230)$, with L = 256 (output range 0–255).

The image contains the pixel values:

r = [30, 70, 90, 110, 140, 200]

Tasks:

a) Derive the equation and slope of each of the three linear segments (0–70, 70–140 and 140–255).

b) Calculate the transformed value for every pixel.

c) Calculate the width of the band 70–140 before and after transformation, and the contrast gain factor.

d) Identify which segments compress the intensity range and which expand it.

e) From the calculated values, assess whether this transformation improves visibility of the diagnostic region and state what is lost in the process.

Q3. Spatial Filtering – Impulse Noise Removal

A 3 × 3 neighbourhood from a grayscale image affected by salt noise is shown below. The centre pixel (255) is a noise impulse.

52	55	58
54	255	57
53	56	59

Tasks:

a) Calculate the output of the 3 × 3 mean (box) filter at the centre pixel.

b) Calculate the output of the 3 × 3 weighted-average filter with mask (1/16) × [1 2 1; 2 4 2; 1 2 1].

c) Calculate the output of the 3 × 3 median filter at the centre pixel.

d) Taking the average of the eight neighbours as the noise-free reference, calculate the absolute error produced by each filter.

e) Based on the numerical results, determine which filter is most appropriate for removing salt-and-pepper noise while preserving edges.

Q4. Edge Detection – Sobel and Prewitt Gradient Analysis

The following 3 × 3 region is taken from the corner of a machined part ($z_1 \ldots z_9$ in row order):

10	10	10
10	10	90
10	90	90

Use the Sobel masks Gx = [−1 0 1; −2 0 2; −1 0 1], Gy = [−1 −2 −1; 0 0 0; 1 2 1] and the Prewitt masks Gx = [−1 0 1; −1 0 1; −1 0 1], Gy = [−1 −1 −1; 0 0 0; 1 1 1].

Tasks:

a) Calculate Gx and Gy at the centre pixel using the Sobel operator.

b) Calculate the gradient magnitude $M = \sqrt{(Gx^2 + Gy^2)}$ and the approximation $M \approx |Gx| + |Gy|$. Find the percentage error of the approximation.

c) Calculate the gradient direction $\alpha = \tan^{-1}(Gy / Gx)$ and state the orientation of the edge.

d) Repeat parts (a) and (b) using the Prewitt operator.

e) If an edge is declared when $M \geq 250$, determine whether each operator detects an edge at this pixel and explain the difference numerically.

Q5. Global Thresholding – Iterative and Otsu Segmentation

A 3-bit image (L = 8) of dark objects on a bright conveyor belt has the following histogram (total 32 pixels):

Intensity r_k	0	1	2	3	4	5	6	7
Frequency n_k	3	6	4	1	1	5	8	4

Tasks:

a) Calculate the global mean intensity m_G.

b) Apply the basic iterative global thresholding algorithm starting from $T_0 = m_G$. Compute the class means m_1 and m_2 and the updated threshold at each iteration until $\Delta T < 0.5$.

c) Using Otsu's method, calculate the between-class variance $\sigma_B^2(k) = P_1(k) \cdot P_2(k) \cdot [m_1(k) - m_2(k)]^2$ for k = 2, 3 and 4 (class C_1 = levels 0…k).

d) Determine the optimum Otsu threshold k*, construct the binary mapping table (0 → background / 1 → object) and calculate the percentage of pixels classified as object.

e) Compare the thresholds obtained by the two methods and state whether the histogram is suitable for global thresholding.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Thresholding and Intensity - Level slicing - PCB Inspection

Given pixel values: $r = [25, 60, 95, 130, 165, 200, 235]$

(a) Binary thresholding

Given transformation:

$$s = \begin{cases} 255, & r \geq 128 \\ 0, & r < 128 \end{cases}$$

Apply the threshold of 128:

Input r	25	60	95	130	165	200	235
Output s	0	0	0	255	255	255	255

Answer: $[0, 0, 0, 255, 255, 255, 255]$

(b) Intensity - level slicing

Pixels btw 90 and 170 are set to 255; all other pixels remain unchanged.

Output: $[25, 160, 255, 255, 255, 200, 235]$

(c) Number and percentage of highlighted pixels

Pixels highlights: 95, 130, 165

Number = 3

Percentage = $3/7 \times 100 = 42.86\%$.

(d) Mean intensity

Original mean: $\bar{r} = \frac{25+60+95+130+165+200+235}{7}$

$$\boxed{\bar{s} = 130}$$

Sliced output mean: $\bar{s} = \frac{25+60+255+255+255+200+235}{7}$

$$\boxed{\bar{s} = 183.57}$$

(e) Comparision and conclusion:

Intensity-level slicing is more appropriate for identifying solder joints because it highlights pixels within the required 90-170 intensity band. Binary thresholding highlights every pixel greater than or equal to 128, including 200 and 235, which are outside the required band.

2) Piecewise-Linear Contrast stretching - X-Ray Image Enhancement.

Given control points (0,0) (70,20) (140,230) and (255,255)

Input pixels: $r = [30, 70, 90, 110, 140, 200]$

(a) Equations and slopes

The slope of each segment is $m = \frac{s_2 - s_1}{r_2 - r_1}$

Segment 1: $0 \le r \le 70$

$$m_1 = \frac{20-0}{70-0} = 0.286$$

$$\boxed{s = 0.286r}$$

Segment 2: $70 \le r \le 140$

$$m_2 = \frac{230-20}{140-70} = 3$$

$$s = 20 + 3(r - 70) = 3r - 190$$

Segment 3: $140 < r \leq 255$

$$m_3 = \frac{255 - 230}{255 - 140} = \frac{25}{115} = 0.2174$$

$$s = 230 + 0.2174(r - 140)$$

The complete transformation is

$$s(r) = \begin{cases} 0.286r, & 0 \leq r \leq 70 \\ 3r - 190, & 70 < r \leq 140 \\ 230 + 0.2174(r - 140), & 140 < r \leq 255 \end{cases}$$

b) Transformed values:

Input r	Formula used	Output s
30	0.286(30)	8.57
70	0.286(70)	20
90	3(90) − 190	80
110	3(110) − 190	140
140	3(140) − 190	230
200	230 + 0.2174(200 − 140)	243.04

(c) Contrast gain factor:

$$V_{original} = 140 - 70 = 70$$

$$V_{transformed} = 230 - 20 = 210$$

$$\text{Contrast gain factor} = \frac{210}{70} = 3$$

d) compression and expansion :-

Segment 1 : Slope 0.286 < 1 → compresses intensity

Segment 2 : Slope 3 > 1 → expands intensity.

Segment 3 : Slope 0.2174 < 1 → compresses intensity.

3) Spacial Filtering – Impulse Noise Removal :

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

(a) 3×3 mean filter :

$$\text{Mean} = \frac{\text{Sum of all pixels}}{9} = \frac{52+55+58+54+255+57+53+56+59}{9}$$

$$= \frac{699}{9} = 77.67$$

(b) Weighted - avg filter :

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Applying weights : $\frac{222+444+1020}{16}$

$$= \frac{1686}{16} = 105.375$$

(c) Median filter : Arrange pixels in ascending order

52, 53, 54, 55, 56, 57, 58, 59, 255

Median = 56

(d) Absolute error of each filter :

$$R = \frac{52 + 55 + 58 + 54 + 57 + 53 + 56 + 59}{8} = \frac{444}{8} = 55.5$$

Absolute error formula: $E = |\text{Filter Output} - R|$

Filter	Output	Absolute Error
Mean	77.67	$\|77.67 - 55.5\| = 22.17$
Weighted average	105.375	$\|105.375 - 55.5\| = 49.875$
Median	56	$\|56 - 55.5\| = 0.5$

4) Edge Detection – Sobel and Prewitt Gradient Analysis

a) Sobel operator

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For $G_x = 0 + [-20 + 180) + (-10 + 90)$

$G_x = 240$

For $G_y = (-10 - 20 - 10) + (10 + 180 + 90)$

$G_y = 240$

b) Gradient magnitude and approximation

$$M = \sqrt{G_x^2 + G_y^2} = \sqrt{240^2 + 240^2} = 339.41$$

$$M = |G_x| + |G_y|$$

$$= 240 + 240 = 480$$

Percentage error (E) = 41.42%

..., / = 3.5238

c) Gradient direction :-

$$\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right) = \tan^{-1}\left(\frac{240}{240}\right)$$

$$\boxed{\alpha = 45^\circ}$$

∴ Indicates diagonal intensity change.

d) Prewitt operator:

$$G_x = 160, \quad G_y = 160$$

Magnitude :- $M = \sqrt{160^2 + 160^2} = 226.27$

Approximation $= |160| + |160| = 320$

Percentage error $= E = \dfrac{320 - 226.27}{226.27} \times 100$

$$E = 41.42\%.$$

e) Edge detection comparision:

$M \geq 250$

Operator	Exact Magnitude	Threshold	Edge detection
Sobel	339.41	250	Yes
Prewitt	226.27	250	No

5) Global thresholding – Iterative and otsu Segmentation:

Given histogram

Intensity r_k	0	1	2	3	4	5	6	7
Frequency n_k	3	6	4	1	1	5	8	4

Total no. of pixels (N) = 3+6+4+1+1+5+8+4 = 32

(a) Global mean intensity:

$$\text{Formula} = m_G = \frac{\sum r_k n_k}{N}$$

$$m_G = \frac{0(3)+1(6)+2(4)+3(1)+4(1)+5(5)+6(8)+7(4)}{32}$$

$$m_G = \frac{122}{32} = 3.8125$$

b) Iterative global thresholding:

Initial threshold : $T_0 = m_G = 3.8125$

$$m_1 = \frac{0(3)+1(6)+2(4)+3(1)}{3+6+4+1} = \frac{17}{14}$$

Class 2 contains intensities 4,5,6,7

$$m_2 = \frac{4(1)+5(5)+6(8)+7(4)}{1+5+8+4} = \frac{105}{18}$$

Updated threshold : $T_1 = \frac{m_1+m_2}{2} = \frac{1.2143+5.833}{2}$

$$T_1 = 3.5238$$

Iteration 2:

As $T_1 = 3.5238$, the two classes remain 0-3 and 4-7. Therefore, the class means remain unchanged

$$T_2 = \frac{1.2143+5.833}{2} = 3.52$$

$$\Delta T = |T_2 - T_1| = 0$$

Since, $\Delta T < 0.5$, the algorithm stops.

Final iteration threshold, $T = 3.5238$

c) Otsu's Method:

$$\sigma_B^2(k) = P_1(k) P_2(k) [m_1(k) - m_2(k)]^2$$

C_1 contains levels 0 to K and C_2 contains levels K+1 to 7.

K	P_1	P_2	m_1	m_2	$\sigma_B^2(k)$
2	13/32	19/32	1.0769	5.6842	5.1202
3	14/32	18/32	1.2143	5.8333	5.2506
4	15/32	17/32	1.400	5.9412	5.1354

d) Optimum Otsu' threshold & binary mapping :-

Maximum between class variance is 5.2506

K=3

$k^* = 3$

Input intensity	Output	Classification
0	0	Background
1	0	Background
2	0	Background
3	0	Background
4	1	Obj
5	1	Obj
6	1	Obj
7	1	Obj